

# ELENA RAIKOVA

## Senior AI and Machine Learning Engineer

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13 years in production software engineering, including 5 years in end-to-end AI and ML systems.  
Classical ML, deep learning, LLM/RAG evaluation, demand forecasting, agentic AI.

## EXPERIENCE

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### DataArt — Senior AI/ML Engineer / Technical Lead

2013 – Present. Remote / Buenos Aires

*Most recent projects:*

#### Failure-attribution evaluation for RAG & agents · *architecture owner*

- Reusable evaluation framework that turns a pass/fail score into a diagnosis — pinpointing which stage failed with transition matrices across runs
- Scored semantic quality with an LLM judge calibrated against human labels
- Verified diagnoses causally by perturbing the suspected component and checking whether the failure changed as expected

#### Demand forecasting + causal price optimization

- Forecasted demand across 140+ parking facilities × 4 targets; benchmarked Chronos-2, Chronos + LightGBM, Prophet, LSTM, DeepAR, TFT. Reduced MAPE from ~50% to < 15% on a 30-day horizon
- Calendar/event features, lag/rolling features, cross-facility features, booking curves, missing-data handling, leakage-safe time splits
- Multivariate, global training across facilities and targets, with transfer to cold-start markets
- Price elasticity under endogeneity via IV/2SLS, regression discontinuity, control function, and panel fixed effects
- Revenue-maximizing price from a monotone-constrained LightGBM demand model pinned to the causal elasticity; served via FastAPI over reproducible DVC pipelines.

#### Agentic AI workflow automation · *architecture owner*

- Agentic workflow automation for a manual royalty-approval process (20k requests/year): LLM-assisted agents embedded in a Power Automate workflow to extract, normalize, and validate Excel/PDF requests, with decision traces shown in Teams adaptive cards.
- Separated deterministic business rules from LLM-assisted decisions, with confidence-based abstention, human approval, sanctions checks, subscription/price-book validation, idempotent processing, malformed-document handling, and full audit trails

#### Enterprise helpdesk Teams chatbot

- MS Teams chatbot for a 6k-user corporate knowledge base: answered user questions and routed requests to the correct helpdesk category using classification methods
- Built a one-pass ModernBERT decision classifier that scores arbitrary answer sets as parallel isolated branches; redesigned the attention mask to prevent cross-option leakage, improving accuracy from ~55% to ~85%.
- Hybrid RAG over ~54K multilingual Confluence docs indexed in Qdrant, GLiNER entity scoping
- Jira Service Management integration: opens tickets from chat, syncs status/comments to the Teams thread

#### Sensor-data lakehouse with real-time ML

- Implemented and evaluated anomaly-detection and forecasting models on sensor data within a Kafka/Dagster/Dremio/S3/MinIO platform
- Benchmarked Chronos-2 against linear and Random Forest baselines

**Earlier engineering (2013–2022).** NLP for trading-form automation and CV parsing/matching; a carrier-grade Ethernet test & diagnostics platform with a Java engine and Python device connector over SSH/TL1/SOAP; Django CRM and an Open edX LMS. Stack spanned Python, Java, JavaScript, Scala, SQL, Docker, Kubernetes.

### ML notes with runnable experiments and interactive visualizations

- **LLMs:** rule induction and LLM-guided program search; calibrated decisions with RL (RLCR); post-training with SFT, LoRA/QLoRA, RLHF/RLAIF, DPO, GRPO; attention variants (MQA, GQA, MLA); agent architectures and temporal grounding
- **Deep learning:** mechanistic interpretability of GPT-2 small (sparse autoencoders, activation patching, induction heads, steering); knowledge distillation on CIFAR-10; an AlphaZero-style Go player from scratch
- **ML systems:** how GPUs run ML and matrix multiplication; LLM inference on a T4 GPU (prefill vs decode); from a PyTorch call to a CUDA kernel
- **Retrieval:** embeddings from LSA and word2vec to BERT, E5 and BGE-M3; chunking strategies; text clustering with UMAP and HDBSCAN
- **Forecasting & causal inference:** ARIMA/SARIMAX, LSTM, TCN, anomaly detection; endogeneity, IV/2SLS, regression discontinuity, panel fixed effects, pricing elasticity

### SKILLS

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**Languages:** Python, Java, JavaScript, Scala, SQL

**ML / DL:** PyTorch, scikit-learn, deep learning, classical ML, model evaluation, hyperparameter optimization, mechanistic interpretability (SAEs, activation patching), knowledge distillation

**LLM / GenAI:** RAG, agents, LLM-as-a-judge, fine-tuning, LoRA SFT (MLX), rejection-sampling fine-tuning, test-time training, RL with calibration rewards, program synthesis;

**NLP:** spaCy, GLiNER, NLTK, entity extraction, classification, document intelligence

**Forecasting:** Chronos/Chronos-2, LSTM, DeepAR, quantile forecasting, booking curves, price optimization, what-if

**Environment:** Power Automate, Dynamics 365, Dataverse, Azure AI Document Intelligence / AI Search, MS Teams, Jira Service Management, Confluence, Kafka, Dagster/Airflow, Dremio, DVC · PostgreSQL, MongoDB, Qdrant, Snowflake, Redis · FastAPI, Streamlit · Docker/Podman, Kubernetes, MinIO, Terraform, GitHub Actions, Sentry, GPU/CUDA, Kaggle

**Cloud:** Azure, AWS

### EDUCATION

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**M.Sc., Applied Mathematics, Informatics and Mechanics** — Voronezh State University, Russia